ALLIOT C. NAGLE

acnagle@utexas.edu \ www.acnagle.com \ Google Scholar

EDUCATION

University of Texas at Austin Aug. 2022 – Dec. 2027 (Expected)
PhD Electrical Engineering
Advisor: Prof. Hyeji Kim

University of Wisconsin-Madison Sep. 2019 – Dec. 2021
MS Electrical Engineering
Advisor: Prof. Dimitris Papailiopoulos

University of Wisconsin-Madison Sep. 2014 – May 2019
BS Electrical Engineering

RESEARCH INTERESTS

Deep Learning	Large Language Models, Reasoning, Vision-Language Models, Representation learning
Model Efficiency	Methods for Accelerated Training and Inference, Neural Network Sparsity

INDUSTRY EXPERIENCE

Texas Instruments, Dallas, TX - *AI Solutions Intern* June 2025 – Aug. 2025

- Developed node library for next-generation AI accelerator chip in a cross-functional team
- Implemented and optimized PyTorch activation functions (e.g., GELU, ELU, SiLU) and nonlinear operations (e.g., exponential, log, tanh, sigmoid) for custom hardware
- Contributed to the chip's functional design through software-hardware co-design

SELECTED RESEARCH PROJECTS

- **Efficient Early-Exit LLM Reasoning** Aug. 2025 – Present
  - Designed a method to predict when an LLM should stop reasoning during online inference (NeurIPS 2026)
  - Achieved over 50% reduction in generated tokens while outperforming prior work
  - Supervised an EPFL MS thesis training LLMs to exit reasoning early via on-policy self-distillation, with Terminator's predicted exit points informing the teacher model; paper in preparation
- **Improving Self-Consistency Through Soft-Averaging** Aug. 2025 – July 2026
  - Mentored a first-year PhD student on a theoretically-motivated Self-Consistency method that replaces hard votes with soft-averaged LLM probability vectors (ICML 2026 SPIGM Workshop)
  - Achieved up to 4% accuracy gains over Self-Consistency and up to 2% over Self-Certainty
- **Semantic Task-Aware Distributed Source Coding** Oct. 2023 – May 2025
  - Proposed a novel framework for learning a single compressed representation shared between a task-specific network and a Vision-Language Model
  - Demonstrated that joint training improves performance for both models
- **Prompt Compression for LLMs** May 2023 – May 2024
  - Formulated an information-theoretic framework of prompt compression for black-box LLMs
  - Achieved superior accuracy-compression trade-offs compared to prior work using query-aware compression
- **Neural Distributed Source Coding** Sep. 2022 – Sep. 2023
  - Built Neural DSC framework achieving state-of-the-art PSNR on stereo image compression with $4-7\times$ fewer parameters than baselines
  - Established theoretical connection between distributed source coding and modified ELBO for asymmetric VAE architectures

- **fMRI-Informed High-Dimensional Brain Network Estimation** Jan. 2021 – July 2022
  - Proposed a weighted group LASSO method for estimating high-dimensional MVAR brain network models from iEEG using fMRI functional connectivity as prior information
  - Reduced data requirements by 2× over prior group LASSO while producing sparser, more accurate connectivity estimates

SELECTED PUBLICATIONS & PREPRINTS

- **A. Nagle**, J. Saydaliev, D. Garbaya, M. Gastpar, A. Makkuva, H. Kim. "Terminator: Learning Optimal Exit Points for Early Stopping in Chain-of-Thought Reasoning," **NeurIPS 2026.**
- R. V. Pandian, **A. Nagle**, H. Kim. "SASC: Soft-Averaged Self-Consistency to Improve Chain-of-Thought Reasoning in Instruct-LLMs," **ICML 2026 SPIGM Workshop.**
- A. Makkuva, M. Bondaschi, A. Girish, **A. Nagle**, M. Jaggi, H. Kim, M. Gastpar. "Attention with Markov: A Curious Case of Single-layer Transformers," **ICLR 2025 spotlight.**
- **A. Nagle***, A. Girish*, M. Bondaschi, M. Gastpar, A. Makkuva, H. Kim. "Fundamental Limits of Prompt Compression: A Rate-Distortion Framework for Black-Box Language Models," **ICML 2024 TF2M Workshop Oral, and NeurIPS 2024 Main Track.**
- A. Makkuva, M. Bondaschi, C. Ekbote, A. Girish, **A. Nagle**, H. Kim, M. Gastpar. "Local to Global: Learning Dynamics and Effect of Initialization for Transformers," **NeurIPS 2024.**
- **A. Nagle***, J. Whang*, A. Acharya, H. Kim, A. G. Dimakis. "Neural Distributed Source Coding," **Special Issue JSAIT 2024.**
- **A. Nagle**, J. P. Gerrelts, B. M. Krause, A. D. Boes, J. E. Bruss, K. V. Nourski, M. I. Banks, B. Van Veen. "Improved High-Dimensional Multivariate Autoregressive Model Estimation of Human Electrophysiological Data Using fMRI Priors," **NeuroImage 2023.**
- K. Sreenivasan, J. Sohn, L. Yang, M. Grinde, **A. Nagle**, H. Wang, E. Xing, K. Lee, D. Papailiopoulos. "Rare Gems: Finding Lottery Tickets at Initialization," **NeurIPS 2022.**
- A. Pensia, S. Rajput, **A. Nagle**, H. Vishwakarma, D. Papailiopoulos. "Optimal Lottery Tickets via SubsetSum: Logarithmic Over-Parameterization is Sufficient," **NeurIPS 2020 spotlight.**

*Equal contribution.

TECHNICAL SKILLS

Programming Languages	Python, C/C++ (CUDA, OpenMP, MPI), MATLAB, Java, Julia
Software & Tools	Amazon EC2, Slurm, HTCondor, Docker, Git, LaTeX, Altium Designer

AWARDS

- ECE Gerald Holdridge Outstanding Teaching Assistant Award — Dec. 2019
- Cockrell School of Engineering Multi-Year Fellowship — Aug. 2022 – Aug. 2026
- Charles W. and Margaret A. Tolbert Endowed Fellowship — Aug. 2026 – Aug. 2027
- ICML 2026 Gold Reviewer — July 2026

TEACHING AND MENTORING

- **MS Thesis Supervisor**, Jakhongir Saydaliev, EPFL — Mar. 2026 – Aug. 2026
- **Research Mentor**, Ruban Vishnu Pandian (PhD student), UT Austin — Aug. 2025 – July 2026
- **Intro to Random Signal Analysis and Statistics**, UW-Madison — Sep. 2019 – May 2021
- **Signals, Information, and Computation**, UW-Madison — Sep. 2018 – May 2019

ACADEMIC SERVICE AND OUTREACH

- **Reviewer**: NeurIPS, ICML, IEEE ISIT
- **Outreach**: Edison Lecture Series GenAI Demo for grades 6–12